

Statistical Arbitrage in Crypto Pair Spreads

Working Draft and Paper Outline, Not a Final Manuscript

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Draft status. This document is a working paper outline prepared from the current project results. It is not a final paper, not a polished submission, and not a claim that all analyses are complete. Several claims below are intentionally phrased as provisional because the project still needs advisor review, cross-exchange validation, microstructure-aware costs, and pre-registered out-of-sample evaluation.

Abstract

This draft summarizes a Math 199 research project on statistical arbitrage in hourly Binance.US USDT crypto pair spreads. The central methodological result is that static ordinary least squares hedge ratios fail to produce stable out-of-sample spreads for most pairs, while a train-only maximum-likelihood Kalman dynamic hedge ratio produces out-of-sample stationary residuals for nearly the entire screened liquidity universe. Downstream trading results are more cautious: machine learning models and z-score rules identify pre-cost mean-reversion signals, but realistic transaction costs and selection-bias corrections sharply weaken the profitability claim. At this draft stage, the defensible conclusion is not that the strategy is tradable, but that dynamic hedge estimation reveals strong non-stationary relative-price structure whose economic value remains unresolved.

How to Read This Draft

This file is an outline-style manuscript. It is meant to collect the current results in paper form and to make the argument easy to revise. Sections marked as provisional should be checked before any final submission. Numerical values come from the repository results documents and scripts as of the current project state, especially `docs/RESULTS.md` and `docs/METHODS.md`.

1 Research Question and Motivation

This project asks whether cryptocurrency prices exhibit stable, quantifiable relative-value relationships, and whether deviations from those relationships tend to mean-revert. The research follows the broad pairs-trading framework of Gatev, Goetzmann, and Rouwenhorst (2006), adapted to a continuously traded crypto market with a changing universe of listed assets.

The project has two distinct goals. The first is statistical: determine whether coin-like relationships exist among liquid crypto assets after avoiding obvious look-ahead bias. The second is economic: test whether any detected mean-reversion signal remains useful after walk-forward validation, transaction costs, and selection-bias correction. The current evidence supports the statistical goal much more strongly than the economic one.

2 Data and Universe Construction

The data set contains hourly OHLCV observations from the Binance.US REST API for USDT spot pairs. Each symbol is stored locally as a parquet file. The current snapshot ends on 2026-01-22. The primary analysis starts at $t_0 = 2024-01-01$, uses a liquidity-filtered universe, and evaluates pairs with walk-forward train/test splits.

The investible universe is constructed as of each analysis time. A symbol is eligible only if its first observation is at least the required minimum-history window before t_0 , and its last observation reaches the start of the analysis period. This first-observation rule was added after an audit found that post- t_0 listings could otherwise leak into the historical universe.

Survivorship remains a limitation. The local store contains currently listed symbols, and public Binance.US delisting information implies that roughly 3–6 liquidity-filtered candidates may be missing during the analysis window. The estimated survivorship-bias bound is about 6.7% as an upper bound and about 5% as a realistic estimate. This does not invalidate the current results, but it should be treated as a material caveat in the final paper.

3 Methods

3.1 Static and Dynamic Spreads

The baseline spread uses a static OLS hedge ratio:

$$s_t = \log(P_{a,t}) - \alpha - \beta \log(P_{b,t}),$$

where α and β are fit on the training window.

The main methodological alternative is a two-state Kalman random-walk hedge model:

$$\begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} = \begin{bmatrix} \alpha_{t-1} \\ \beta_{t-1} \end{bmatrix} + w_t, \quad \log(P_{a,t}) = [1 \quad \log(P_{b,t})] \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} + v_t.$$

The process and observation noise parameters are fit by maximum likelihood on training data only. For test data, parameters are held fixed and the filter is rolled forward from the final training state. This distinction is important: the reported Kalman result is not an in-sample whitening artifact.

3.2 Prediction Models

The project compares simple baselines, a z-score rule, gradient boosting, an LSTM, and transformer variants. Models use backward-looking spread, volume, correlation, volatility, and market-context features. Labels are three-class: revert, persist, or diverge over a 24-hour horizon. The headline walk-forward run uses 27 splits with 90 days of training, 30 days of testing, and a 30-day step.

3.3 Backtesting and Statistical Corrections

Trading evaluation uses a state-machine entry/exit rule: open when the model predicts reversion and $|z| \geq 2$, close when $|z| \leq 0.5$, the model predicts divergence, or the spread changes sign. Transaction-cost experiments include zero cost, 5 basis points round-trip per leg, and 15 basis points round-trip per leg.

The project also reports Newey-West HAC corrections because hourly labels have overlapping 24-hour targets, and deflated Sharpe ratios because many model and strategy configurations were tried. These corrections are central to the current cautious interpretation.

4 Current Results

4.1 Result 1: Static OLS Fails Out-of-Sample

Static hedge ratios do not produce robust out-of-sample stationary residuals. Among the top 10 selected pairs, only 2 maintain out-of-sample ADF $p < 0.05$. Across the full 1,219-pair liquidity-screened universe, static OLS passes $p < 0.05$ for only 66 pairs, or 5.4%.

Table 1: Out-of-sample stationarity screen, static OLS versus Kalman dynamic hedge.

ADF threshold	Static OLS passes	Kalman passes
$p < 0.001$	0 (0.0%)	1,174 (96.3%)
$p < 0.005$	1 (0.1%)	1,188 (97.5%)
$p < 0.010$	10 (0.8%)	1,198 (98.3%)
$p < 0.050$	66 (5.4%)	1,215 (99.7%)
$p < 0.100$	142 (11.6%)	1,217 (99.8%)

4.2 Result 2: Kalman Dynamic Hedge Ratios Recover Stationarity

The strongest current finding is that dynamic hedge ratios recover stationary-looking residuals for nearly all liquid pairs under an honest train/test protocol. On the full 1,219-pair screen, 99.7% of pairs pass $p < 0.05$, and 96.3% pass $p < 0.001$. The selected top-10 pairs all pass $p < 0.001$ under the Kalman model.

This should be the leading result in the final paper unless advisor review identifies a flaw in the protocol. The interpretation is that crypto pair relationships are real but non-stationary: a fixed hedge ratio is usually too rigid, while a slowly drifting hedge ratio captures the relative-price structure.

4.3 Result 3: ML Helps, but the Main Signal Is Still Spread Z-Score

In the post-audit 27-split walk-forward run, the LSTM has the strongest `pnl_mean_to_std`, followed by the booster and z-score rule. The small parameter-matched transformer is not competitive in this run, although a larger transformer later performs much better on a related pre-audit data set.

Table 2: Post-audit walk-forward model comparison. Values are current draft results.

Model	<code>pnl_mean_to_std</code>	Win rate	Trades/split
LSTM	0.376	0.795	2,083
Booster	0.356	0.741	2,355
Z-score	0.282	0.963	588
Majority	0.021	0.028	185
Small transformer	0.018	0.437	691
Random	0.001	0.502	2,689

A feature ablation suggests that the learned models are mostly refining a classical z-score signal. Dropping `latest_spread_z` reduces accuracy by 14 percentage points, macro F1 by 19 percentage

points, and PnL by roughly 79%. Other features have much smaller effects in the current single-split ablation. This is an important draft conclusion: the ML models do not yet demonstrate a qualitatively new source of alpha beyond the spread deviation itself.

4.4 Result 4: Cost and Selection Bias Substantially Weaken Trading Claims

The strategy looks strong before costs, with annualized Sharpe ratios around 8 under the state-machine backtest. That headline does not survive careful interpretation. At 5 bps round-trip per leg, Sharpe falls to roughly 0.8–0.9. At 15 bps round-trip per leg, the strategy is deeply negative.

Table 3: Cost sensitivity under the state-machine backtest.

Cost per leg	LSTM Sharpe	Z-score Sharpe	Booster Sharpe
0 bps	8.08	8.03	8.27
5 bps	0.80	0.83	0.90
15 bps	-14.04	-13.88	-14.17

Deflated Sharpe correction further weakens the profitability claim because the project tried many strategy configurations. At a modest trial count, the pre-cost result may remain reportable. At a conservative count around 50 trials, even the pre-cost result is not clearly distinguishable from the chance-best result. The 5 bps result does not survive deflated-Sharpe correction at any plausible trial count. The current paper should therefore retract any break-even cost claim.

5 Negative Results

The HMM regime filter is a robust negative result. In post-audit runs, the filter suppresses 40–55% of bars and reduces total PnL across models rather than isolating productive mean-reversion regimes. This remains true under 2-state and 3-state variants and across static and Kalman pipelines.

Single-split evaluation is also unreliable. An earlier single chronological split overstated the transformer result. The 27-split walk-forward evaluation reduced the small transformer’s apparent performance until its confidence interval included zero. This supports treating walk-forward evaluation with uncertainty intervals as the minimum reporting standard for this project.

6 Audit Findings

The project found and fixed several methodological issues during review:

- post- t_0 listings could enter the historical universe.
- an in-sample Kalman ADF check was circular and was replaced by a train/test protocol.
- per-pair label thresholds could use information beyond the relevant training boundary.
- the Kalman spread construction had a train/test parameter discontinuity.
- the backtest Sharpe denominator could be distorted by changing active capital.
- `time_since_zero_crossing` was confirmed as a soft leakage feature and removed from the post-audit interpretation.

The repository currently includes leakage and backtester sanity tests. The draft should report these as evidence of methodological cleanup, not as proof that all remaining bias has been eliminated.

7 Limitations

This draft should retain the following limitations prominently:

- Results use one exchange, Binance.US.
- Results use one quote currency, USDT.
- Results use one bar interval, hourly.
- Survivorship bias remains because delisted assets are missing from the local store.
- The backtester uses flat slippage and does not model market impact.
- Spot shorting and funding constraints are not fully modeled.
- All main results use $t_0 = 2024-01-01$. Start-date sensitivity remains pending.
- Deep-learning results are mostly single-seed.
- No pre-registered final strategy has been evaluated exactly once.

8 Provisional Discussion

The current results suggest a distinction between statistical structure and tradable alpha. Dynamically hedged crypto pair spreads appear highly mean-reverting in a statistical sense, especially compared with static OLS spreads. However, converting that structure into a realistic trading strategy is much harder. Transaction costs, selection bias, and market microstructure could plausibly absorb the apparent edge.

This is still a useful result. A final paper can argue that the appropriate model for crypto pair relationships is time-varying rather than static, while also showing that statistical mean reversion is not sufficient evidence of profitable arbitrage. That framing is more defensible than claiming a finished profitable strategy.

9 Work Needed Before Final Submission

Before this becomes a final manuscript, the project should ideally complete:

1. advisor review of the Kalman OOS protocol.
2. a rerun of downstream pair selection using Kalman-screened pairs rather than correlation fallback.
3. cross-exchange validation on Coinbase or Kraken.
4. L2-data ingestion and market-impact modeling.
5. multi-seed deep-learning runs.

6. start-date sensitivity tests.
7. one pre-registered model configuration evaluated once.

10 Draft Conclusion

At this stage, the project supports a cautious conclusion. Static OLS hedge ratios are inadequate for hourly Binance.US crypto pair spreads, while Kalman dynamic hedge ratios recover strong out-of-sample stationary residuals across nearly the entire liquid universe. Predictive models can exploit this signal before costs, but the resulting trading performance does not yet survive realistic cost assumptions and selection-bias correction strongly enough to claim deployable arbitrage. The final paper should emphasize the dynamic hedging result and present the trading results as preliminary, conditional, and limited by current data and execution assumptions.

References

- [1] Gatev, E., Goetzmann, W. N., and Rouwenhorst, K. G. (2006). *Pairs Trading: Performance of a Relative-Value Arbitrage Rule*. *Review of Financial Studies*, 19(3), 797–827.
- [2] Bailey, D. H., and Lopez de Prado, M. (2014). *The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality*.